Geometry Review: Unit 7 — Area, Surface Area & Volume

Answer Key

Station 1: Area of Polygons and Composite Figures

1. 126 cm^2
2. 35 in^2
3. 60 m^2
4. 66 $\frac{1}{2}(8+14)(6)$
5. 80 $\frac{1}{2}d_1d_2$
6. 54; 36 *hypotenuse* $= 15$
7. 87
8. 58
9. $60+\frac{9}{2}\pi \approx 74.1$
10. 6 $48=\frac{1}{2}(16)h$

Station 2: Surface Area of Prisms and Cylinders

1. 150 $6\cdot 5^2$
2. 248 $2(24+40+60)$
3. 132 $2(6)+(3+4+5)(10)$
4. 66π $2\pi(3)(8)+2\pi(9)$
5. 120π
6. 120π $r=5$*:* $70\pi+50\pi$
7. 6
8. 118 $40+2(24)+2(15)$
9. $80\pi \approx 251.3$
10. 4 $2\pi(2)h+2\pi(4)=24\pi$

Station 3: Volume of Prisms and Cylinders

1. 105
2. 216
3. 120 $\frac{1}{2}(6)(4)\cdot 10$
4. 144π
5. 180π $r=6$
6. 6
7. 3
8. 4608 in^3
9. $400\pi \approx 1256.6\text{ cm}^3$
10. 8 2^3

Station 4: Pyramids, Cones, and Spheres

1. 120 $\frac{1}{3}(36)(10)$
2. 120
3. $V=12\pi;\ SA=24\pi$ *slant height* $=5$*;* $\pi(3)(3+5)$
4. 10; 60π; 96π
5. $V=36\pi;\ SA=36\pi$
6. $V=288\pi;\ SA=144\pi$
7. $r=5;\ V=\frac{500\pi}{3}$
8. $\frac{128\pi}{3}$
9. 4 $\frac{1}{3}\pi r^2(12)=64\pi$
10. 48π $30\pi+18\pi$